

Why all this sex?

Ayla examined her son again, trying to remember the reflection of herself. My forehead bulges out like that, she thought, reaching up to touch her face. And that bone under his mouth, I've got one too. But, he's got brow ridges, and I don't. Clan people have brow ridges. If I'm different, why shouldn't my baby be different? He should look like me, shouldn't he? He does a little, but he looks a little like Clan babies, too. He looks like both. I wasn't born to the Clan, but my baby was, only he looks like me and them, like both mixed together.

(from *The Clan of the Cave Bear* by Jean M. Auel)

In the extract above, Ayla realised that her baby was a mixture of two races. The mixing of races is an example of how variety in a population is generated.

Look at the dogs below. What differences can you see? How did this variation come about when all the individuals belong to the same species?

A good variation

Sexual reproduction brings together the genetic material from an organism's own parents in a great variety of new combinations. A result of this can be seen in your own family. You

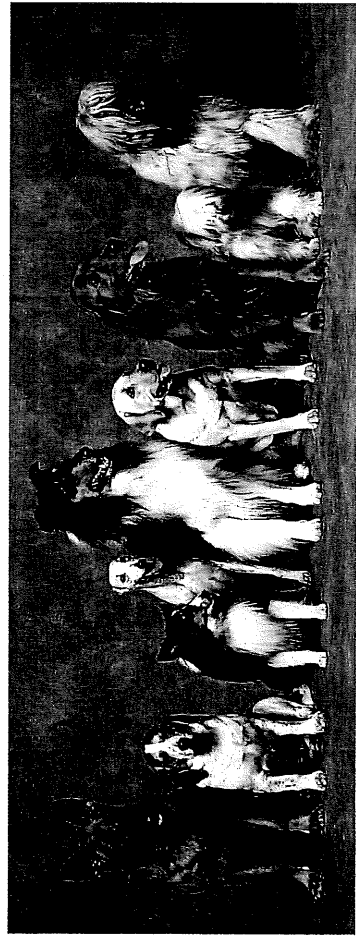
have a different combination of your parents' genetic material to any of your sisters or brothers. The variety that exists in a population of organisms is often a great biological advantage. In the long term, it allows a species to withstand changes in the environment. Although an environmental change may result in the death of many individuals, others with slightly different characteristics may be able to survive it, allowing the species to survive.

Variation also has short-term advantages for a species. The likelihood that an epidemic disease will wipe out a population is reduced by variation

being present in a population. This is because the individuals of the population will be susceptible to a disease in different ways, making it much more difficult for a parasite to make a rapid killing. Overall, it is important to appreciate that the environment in which a species lives is not uniform — it is made up of many different micro-environments. Therefore, the best strategy for a species to survive is to possess a variety of types that can live efficiently in different ways.

Variety through sex

Each of the cells in your body contains genetic information in structures called **chromosomes** from your mother (maternal chromosomes) and your father (paternal chromosomes). When you start producing sex cells or **gametes** (sperm or ova), an opportunity is provided to mix up these paternal and maternal chromosomes.



Different breeds of the species *Canis familiaris*



Our gametes contain a mixture of genetic information from our mother and father.

The price to pay

Though sex is a source of variation, and variation can be valuable to a species, there are prices to be paid. Sexual reproduction is generally more complicated and risky than asexual reproduction. A flowering plant, for instance, has to invest a lot of energy to produce flowers, and possibly nectar too. When animals mate, they have to forgo opportunities to feed, and they make themselves conspicuous which makes them easier prey for their enemies. Complex animals put a lot of energy into caring for their young. Sex is only worthwhile if its benefits outweigh the costs and the prevalence of sexual reproduction in both animals and plants show that they usually do.

Best of both

Many species can reproduce by both asexual and sexual means. When the environmental conditions are favourable, they reproduce rapidly asexually. However, when times are hard or when the environment is changing, they switch to the sexual option, seemingly in order to reap the advantages in variation that are required in adverse environmental conditions.

Activities

Remember

List some of the advantages and disadvantages of sexual reproduction.

Think

1. Asexual reproduction produces no variation. Explain why this is risky for the continuation of a species.
2. Apply the statements 'Variety is the spice of life' and 'Opposites attract' to the principles of sexual reproduction and variation in organisms.

SCIFACTS

There is very little genetic variation among the present day cheetah population. Scientists believe that this is because at some time in the past all cheetahs were wiped out except for one mating pair. The genes of this pair are now found in all cheetahs.

Getting together . . .

Let's talk about sex. Sexual reproduction is not something to be embarrassed about, but something incredible and fascinating.

Sexual reproduction occurs when offspring result from the joining together (**fusion**) of a male reproductive cell and a female reproductive cell. These special reproductive cells are called **gametes** and they are produced in the reproductive organs of the organism. In animals, male gametes are called **sperm** whereas female gametes are often referred to as **ova** (singular: ovum) or **egg** cells.

Reproductive systems are designed to bring the male and female gametes together. The joining of sperm and egg cells is called **fertilisation**. Once fertilisation has occurred, the information from the two parents is mixed. This results in the production of offspring that are different from their parents. It also means that if there are other offspring, they, too, are different. The combination of a sperm cell and an egg cell is called a **zygote**.

Outside . . .

Gametes are very tiny and dehydrate very easily, so they need special conditions to survive. In some animals, especially those that live or breed in water, like fish and amphibians, fertilisation occurs outside the female's body. This is called **external fertilisation**. In this situation, the female releases her unfertilised eggs into the water to be fertilised by the male's sperm, which are also released into the water. This form of fertilisation is often

considered as being inefficient, because many of the gametes released do not combine and are wasted. Of those that are successful, many of these may be lost in their struggle to survive. They may have to escape predators or dangerous situations, or may have problems obtaining resources such as food and shelter.

... or inside

In animals that live and breed on land, **internal fertilisation** occurs. This keeps the gametes inside the body, so there is no chance of dehydration occurring. In this case, the sperm are introduced into the female by a process called **copulation**. **Mating** or **sexual intercourse** are other terms used to describe this event. The fertilised egg may be retained and develop within the female, as in the case of humans, or it may be enclosed within a protective shell and released, as in birds and many reptiles.

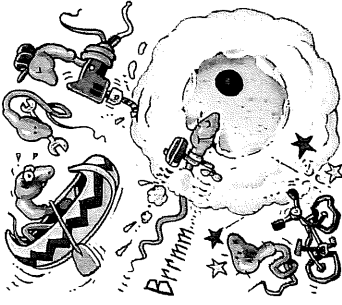
Sperm

In humans, the sperm are mixed with a nutrient-rich fluid before their release, or **ejaculation**, from the male's penis. This combination of fluid and sperm is referred to as **semen**. The sperm make up less than 1 per cent of the semen. An ejaculation may release about 400 million sperm. Sperm are made up of three distinctive parts: the head, the middle section and the tail. Swishing its tail, and powered by the middle section, the sperm swims like a tadpole at about 4 mm per hour to reach its goal — the egg.

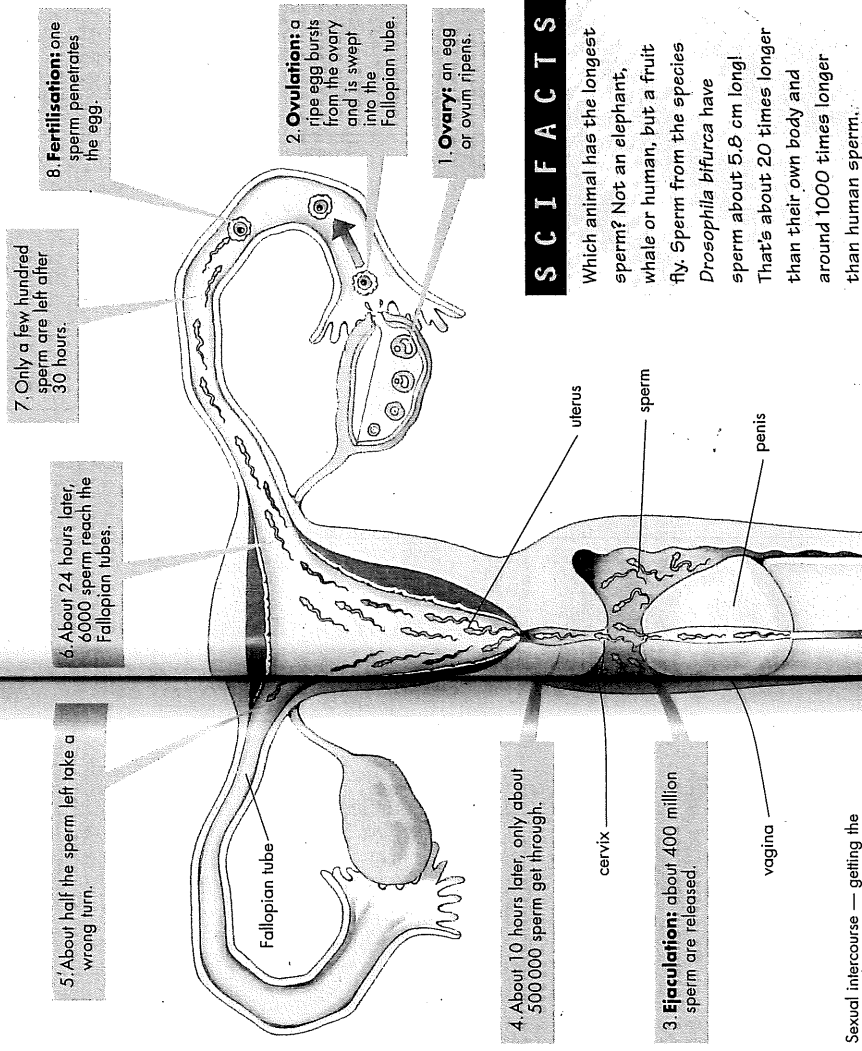
containing the nucleus continues to move towards the nucleus of the ovum to join with it. Because the nucleus contains the information from the parents, this is the point at which the information is mixed together. Once this has occurred, the resulting zygote begins its growth into a new individual. The amazing thing about all of this is that we all started this way. Wow! Imagine having been both a sperm and an egg cell. Isn't that incredible?

Success, at last!

Upon reaching a ripe egg, hundreds of sperm surround it and try to break through. The sperm release an enzyme that dissolves the covering of the egg and enables one of them to penetrate it. Once a sperm has done this, no other can get through. The successful sperm then sheds its body and tail, and the head



The successful sperm fertilises the ovum.



SCIENCE FACTS

Which animal has the longest sperm? Not an elephant, whale or human, but a fruit fly. Sperm from the species *Drosophila bifurca* have sperm about 5.8 cm long! That's about 20 times longer than their own body and around 1000 times longer than human sperm.

Activities

Remember

1. Define the term sexual reproduction.
2. Put the following words into a sentence: gametes, sperm, ovum, zygote, fertilisation.
3. Describe the difference between external and internal fertilisation.
4. Draw labelled diagrams of a sperm and an ovum.
5. Describe the events that take place when the sperm reach a ripe ovum.

Think

1. Hens' eggs are considerably larger than human ova and contain food reserves. Suggest a reason for the differences.
2. Why is internal fertilisation generally more efficient than external fertilisation?
3. Why doesn't fertilisation occur each time a couple have sexual intercourse?

Research

1. Find out more about the gametes of at least four different animals. Display your finding either as models or as diagrams on a poster.
2. There is a theory that by wearing tight jeans, human males may affect the development of their sperm. Find out if there is any scientific evidence to support this theory.

Create

1. Write a story, play or poem about the successful sperm, from ejaculation to when it fertilises the ovum.
2. Design a board game which incorporates information about sperm and eggs.
3. Draw a descriptive timeline which includes: ovulation, ejaculation, sexual intercourse, the various stages of the sperm's travels through the female's reproductive tract, and fertilisation.

Private parts

We've all got private parts. Let's find out where they are and what they do.

Gamete factories...

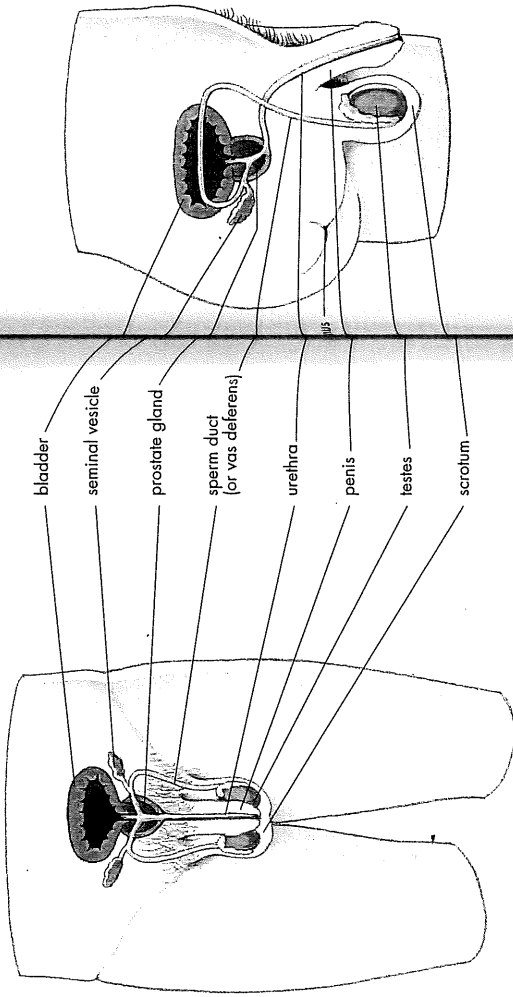
Although testes and ovaries may look different, these two organs have the same job. They both make gametes.

Sperm are made in the **testes** of a male when he is sexually mature. Testes hang from the body within the **scrotum** to maintain sperm at a temperature of about 3°C below that of the rest of the body. This temperature difference is essential for successful sperm production. Tight underwear or jeans can increase the temperature of the testes and so increase the number of damaged sperm produced.

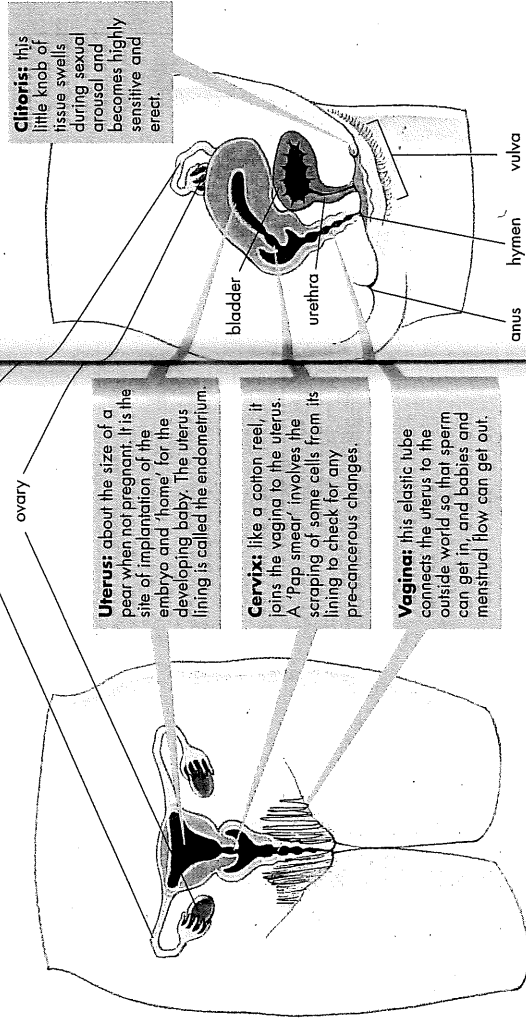
Ova (the female's sex cells) are made in the ovaries. Females are born with about 400 000 eggs, or ova, in their ovaries. These eggs are in sacs called **follicles**. Usually only one ovum (singular for ova) is ripened and released into the **Fallopian tube** (or oviduct) each month.

Fallopian tubes are about the diameter of a human hair. They form a tunnel in which the sperm and ovum meet and are hence the site of **fertilisation**. Damage to these tubes can prevent the sperm and egg from meeting.

If the fertilised egg does not move down into the uterus, but remains in the tube, an ectopic pregnancy may result. If this is not detected and treated before about eight weeks, the tube may rupture and threaten the woman's life.



The male reproductive system



The female reproductive system

Clitoris: this little knob of tissue swells during sexual arousal and becomes highly sensitive and erect.

Uterus: about the size of a pear when not pregnant. It is the site of implantation of the embryo and 'home' for the developing baby. The uterus lining is called the endometrium.

Cervix: like a cotton reel, it joins the vagina to the uterus. A 'Pap smear' involves the scraping of some cells from its lining to check for any pre-cancerous changes.

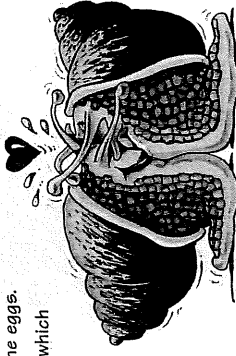
Vagina: this elastic tube connects the uterus to the outside world so that sperm can get in, and babies and menstrual flow can get out.

SCIFACTS

Did you know that not all plants or animals have separate sexes? Many invertebrates are both male and female at once. These interesting combinations are called hermaphrodites. This enables an individual to achieve greater reproductive efficiency than if it was just the one sex.

Snails have been around for 600 million years and have developed intriguing methods of reproduction. Each snail has an organ called an ovotestis which makes both sperm and eggs and a single tube to carry both the sperm and the eggs.

After a complex courtship, in which hermaphrodite snails rear up, each pressing their muscular foot against their partner, and stroking each other with their tentacles, they simultaneously insert their sex organ into the other's body. In this manner, each snail gives sperm to the other and each has its eggs fertilised by the other.



Activities

Remember

1. State how the following pairs are (i) similar and (ii) different:
 - (a) ovaries and testes
 - (b) vas deferens and the Fallopian tubes
 - (c) penis and clitoris
 - (d) seminal vesicles and prostate gland
 - (e) uterus and vagina.
2. Why is tight underwear not recommended for males?
3. What is a 'Pap smear' and why do doctors do it?
4. What is an ectopic pregnancy?

Research

1. (a) Make a model of the reproductive system of a human and another animal of your choice.
(b) How are they different? How are they similar?
2. Research the reproductive system of an animal of your choice. Describe how it reproduces and draw a diagram of its reproductive parts. Present your information in a poster.
3. Find out about the mating flight of the queen bee.

Think

Describe three ways in which animals may increase their chance of successful reproduction by having specialised reproductive structures or techniques.

Imagine

Imagine that you are a sperm or an egg. Find out more about how you are produced, stored and used in sexual reproduction. Write a dramatic story about your life.

Male reproductive system

Male sex cells are called **sperms** and are produced in organs called **testes**. Human testes are oval in shape. They are about 5 cm long and are contained in a protective bag called the **scrotum** (Figs. 1 and 2). Sperms are produced continuously from puberty (sexual maturity) until about seventy years of age. Puberty in males occurs between 10 and 17 years of age.

The inside of a testis is divided into about three hundred compartments each containing up to three narrow twisted tubes (Fig. 3). These tubes are lined with dividing cells which produce sperms. The hundreds of sperm-producing tubes join together and form a smaller number of **collecting ducts**. Sperms move out of the testis through these ducts into a single coiled tube called the **epididymis**. This is a storage area for sperms. Each epididymis leads into a sperm duct, or **vas deferens**. The sperm ducts, one from each testis, rise up the body until they are joined by a duct from a **seminal vesicle**. The sperm ducts join together near the base of the bladder

forming a single tube called the **urethra**. This junction occurs inside the **prostate gland**. Lower down, the urethra is joined by a duct from Cowper's gland and then it leads to the outside of the body through the **penis**.

The walls of the penis contain sponge-like spaces called **erectile tissue**. When a male is sexually stimulated the erectile tissue fills with blood making the penis erect and firm. During sexual intercourse the erect penis is inserted into the vagina of the female and moved back and forth. These movements stimulate sense organs in the penis and eventually cause rhythmic contractions in the walls of the urethra which propel a liquid called **semen** into the female. This event is called an **ejaculation**. Semen consists of millions of sperms floating in a liquid from the seminal vesicles and from the prostate and Cowper's glands. This liquid nourishes the sperms and causes their tails to make the swimming movements which propel them through the female reproductive system to an egg.

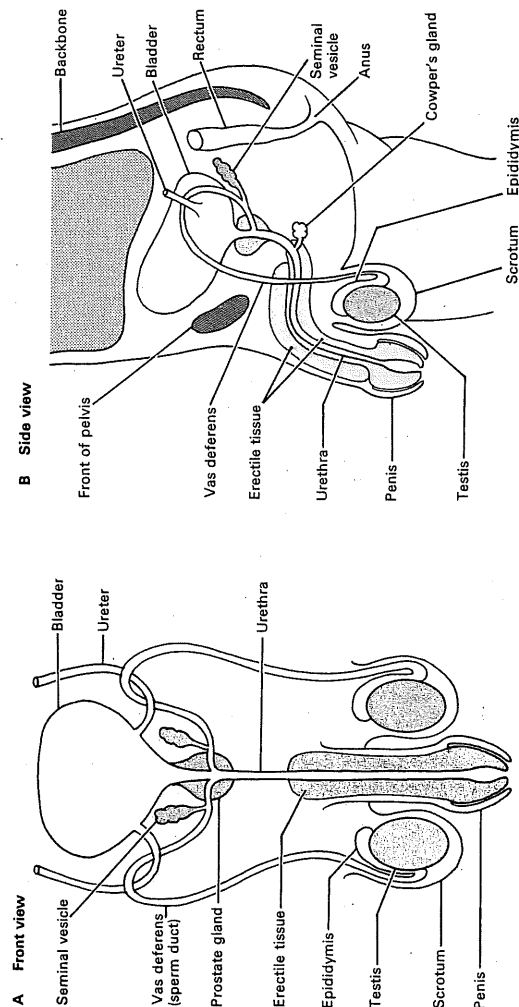


Fig. 1. Diagram of male reproductive system (front view with testes drawn further apart than normal).

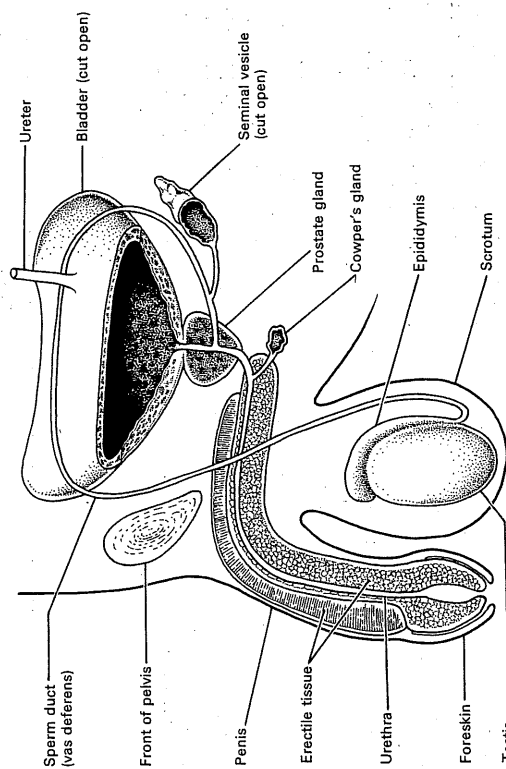


Fig. 2. Male reproductive system (side view). During sexual intercourse the penis becomes erect and is inserted into the vagina of the female. Rhythmic contractions of the sperm ducts propel semen into the female. Semen consists of sperms and liquid from the seminal vesicles, the prostate gland, and Cowper's gland.

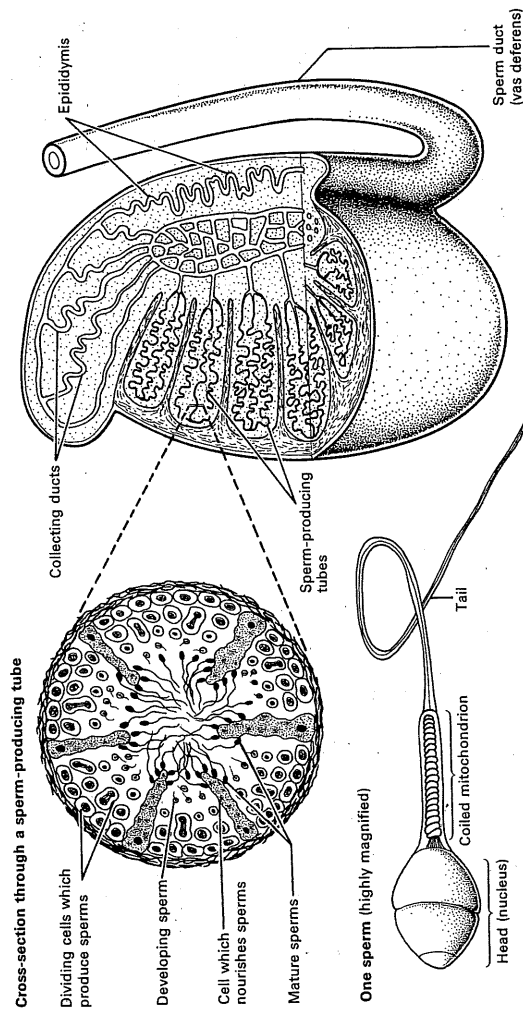


Fig. 3. A testis consists of compartments containing tubes lined with dividing cells which make sperms. Sperms are sex cells. The head of a sperm contains the cell nucleus and this is propelled by the tail.

Female reproductive system

Female sex cells called ova (*singular*: ovum) are produced by two ovaries. Human ovaries are 3 cm long and are attached to the back wall of the abdomen above the kidneys (Fig. 1). Ova begin forming in the ovaries before birth, and a newborn baby girl has a million or more partly formed ova in her ovaries. No new ova develop from this time onwards, however, and during childhood most existing ova disintegrate.

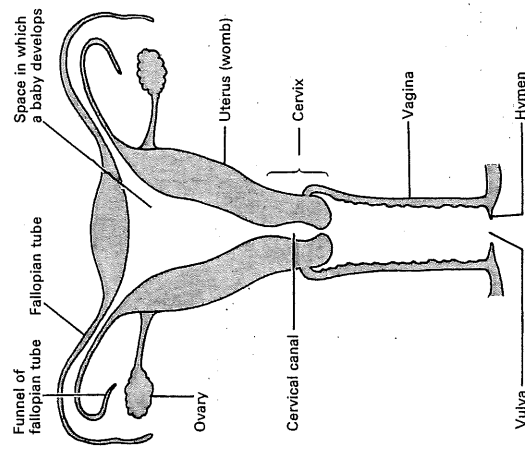
Puberty (sexual maturity) occurs in girls between eight and fifteen years of age, and by this time each ovum is enclosed in a mass of cells called a **follicle**. Once a month the pituitary gland secretes a hormone which causes one of these follicles to absorb liquid, expand, and burst, releasing its ovum from the

ovary. This event is called **ovulation** (Figs. 2 and 3). Ovulation occurs once a month, unless interrupted by pregnancy, until about the age of fifty.

After leaving an ovary the ovum passes into a **fallopian tube**. Each fallopian tube has a funnel-shaped opening and is lined with microscopic hairs called **cilia** (Fig. 2). Peristaltic contractions in the walls of the fallopian tubes sweep an ovum along until it reaches a wider thicker-walled tube called the **womb**, or **uterus**. An ovum which has been fertilized by a sperm develops into a baby in the uterus.

A ring of muscle called the **cervix** closes the lower end of the uterus where it joins another tube, the **vagina**. The vagina extends to the outside of the body where there is an opening called the **vulva**.

A Front view



B Side view

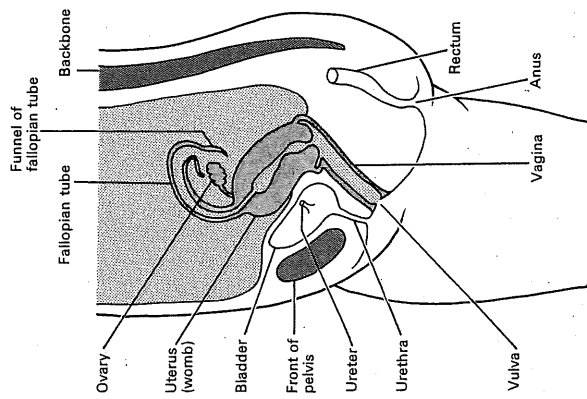


Fig. 1. Diagram of the female reproductive system.

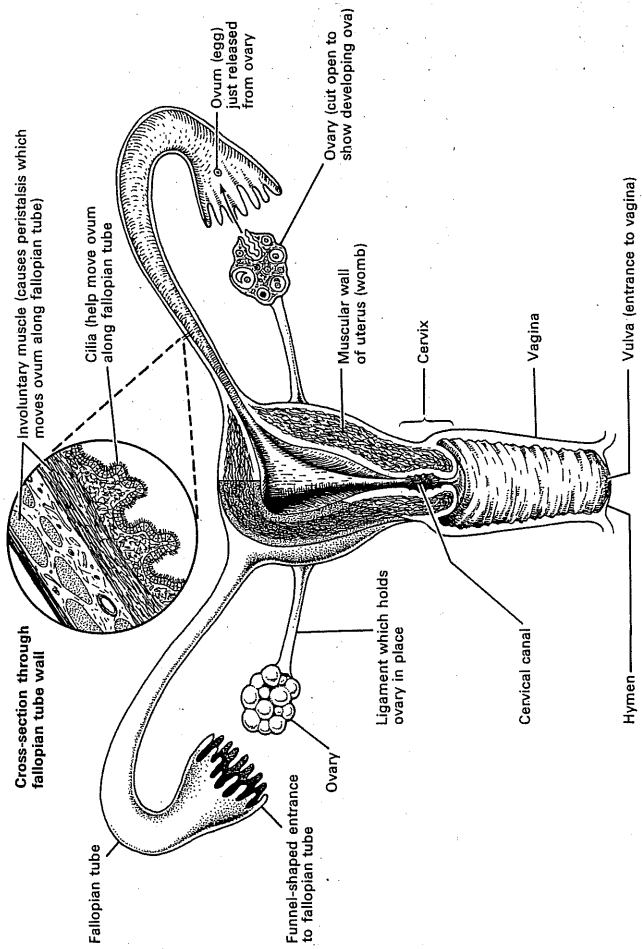
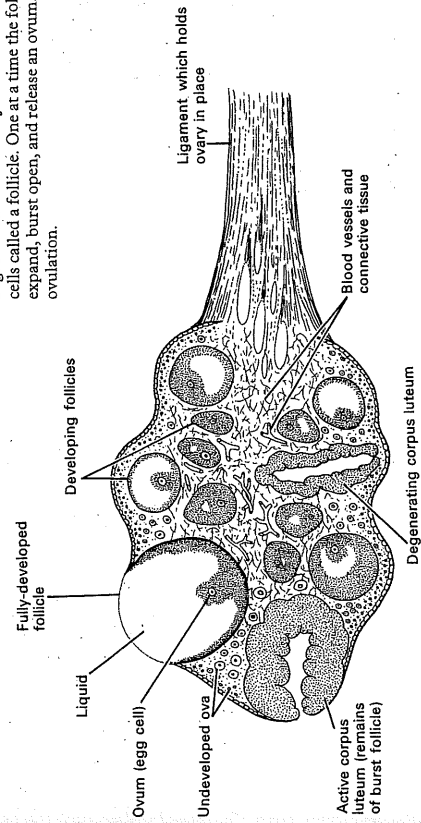


Fig. 2. Female reproductive system (left side cut open). Once a month an ovary releases an ovum. It is moved along a fallopian tube by peristaltic contractions and by the action of cilia.

Fig. 3. Inside an ovary each ovum is enclosed in a cluster of cells called a follicle. One at a time the follicles absorb water, expand, burst open, and release an ovum. This is called ovulation.



Changing times

Perhaps you have noticed that you are changing. Have you noticed any hairs in parts where they weren't before, changes in your body shape or height, changes in your interests ...?

From child to adult

Changes that you may be currently aware of are indications that you are becoming an adult. These physical changes are called **puberty**. The term 'puberty' comes from the Latin word *pubertas*, which means adulthood.

The main purpose of the changes that occur in puberty is to enable you to start producing children. Your sex organs grow and develop. Males begin to produce sperm and females begin to develop the ova they were born with. When combined, these gametes can produce babies.

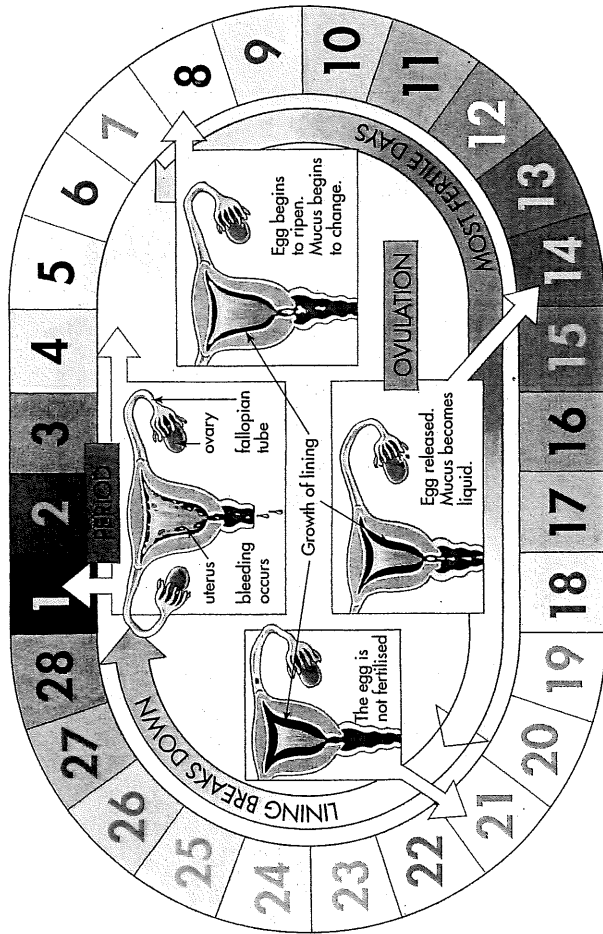
Although all the eggs or ova are present at birth, it is only during puberty that they begin to ripen. Each month, one ovum matures and is released from the ovary into the Fallopian tube.

The release of the ovum is called **ovulation**. This continues about every 28 days until the woman enters menopause.

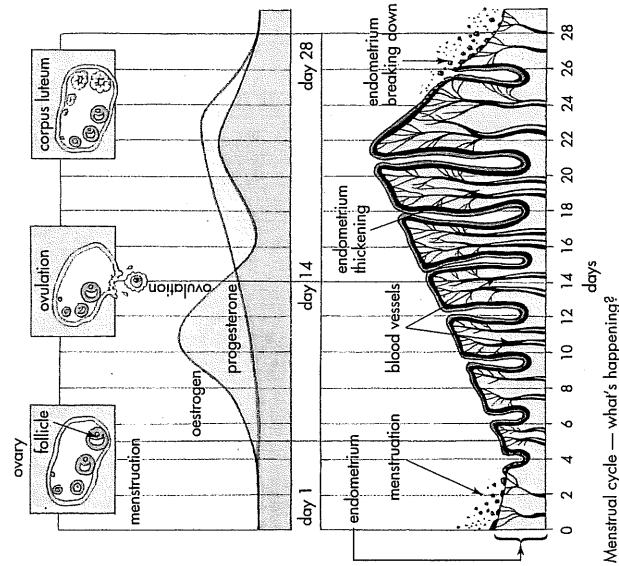
Girls are born with about 400,000 eggs. Only about 400 of these will mature and be released during her life; the others will not mature.

Each month, the lining of the uterus prepares itself for the fertilised egg. If the egg is not fertilised, the uterus lining is shed through the vagina. This monthly discharge, or shedding, is called **menstruation**, or a period. Sometimes there may be some discomfort or abdominal cramps during the first few days of your period. If you are having periods it is a sign that you are also ovulating. This means that you are physically able to get pregnant and have a baby.

Most girls experience their first period between the ages of 11 and 14, although others may have it two years earlier or later. The first menstrual period is called **menarche**. A period generally lasts about four to six days, with varying amounts of discharge over this time. The



Solving the period puzzle — looking for patterns ...



Menstrual cycle — what's happening?

discharge contains cells from the lining of your uterus which was built up in preparation for a baby. Your **menstrual cycle** is the time from the first day of one period to the first day of the next. It is usually about 28 days, but this varies in different women. It is quite common for your menstrual cycle to be irregular at first — until your body settles into its own pattern.

SCIFACTS

The feeling of 'being in love' is not a product of the heart. That happy, dreamy feeling experienced when you 'fall in love' is partly due to a chemical produced in your brain, called phenylethylamine.

Activities

Using data

1. Use the illustration of menstruation, and other information from these pages, to answer the following questions.

- What is ovulation?
 - At what day in a 28-day cycle is ovulation likely to occur?
 - At which time in the cycle is sperm most likely to meet (and fertilise) an egg?
2. Describe the changes in oestrogen and progesterone levels throughout the cycle.
- Which hormone is highest just before ovulation?
 - Which hormone is highest when the uterine lining is the thickest?
 - At which time would the lining provide the best 'home' for a fertilised egg?
3. Translate the graph into a 28-day calendar.

Think

- Suggest why the hips become 'flesher' and the pelvic bones widen in females during puberty. The testes continue making sperm for the rest of the man's life. How is this different from gamete production in a woman? What are the consequences of this?

- If a female has menstrual cycles, is she potentially able to have babies? Explain.

Research

Write down ten questions that you have about puberty or reproduction. Use a variety of texts and resources to find the answers to them. Report the findings to your friends.

Create

- Construct a crossword with your own clues and answers, from information found on these pages.
- Write and act out a play to demonstrate the menstrual cycle. Include the following stages: the egg ripening, ovulation, movement of the unfertilised egg into the Fallopian tube and through the uterus, and both the egg and the uterine lining being shed out through the vagina.
- Get everyone in the class to write down two to five questions about the changes that occur during puberty and reproduction in a 'Dear Ethel' magazine format. After your teacher has collated these, select at least two to investigate. Report your findings by compiling a class 'Dear Ethel' magazine.

The menstrual cycle

Starting at puberty the female reproductive system undergoes a regular sequence of events, repeated monthly, called the **menstrual cycle**. One menstrual cycle lasts about twenty-eight days, although it can be longer or shorter than this.

During one menstrual cycle an ovum is released from an ovary and the lining of the womb (uterus) is renewed. It develops to a stage at which it can receive and nourish the ovum should this be fertilized and begin to grow into a baby. These events are controlled by hormones from the ovaries and from the pituitary gland, and are summarized in Figure 1.

Events during one menstrual cycle

1. From day one to about day five of a cycle the lining of the uterus breaks down into loose tissues which flow out of the vagina together with 50 to 250 cm³ of blood. The technical term for this process is **menstruation**, which is commonly known as 'a period'.
2. When bleeding stops an ovary is stimulated by secretion of a chemical called **follicle-stimulating hormone** from the pituitary gland. The ovary responds to this hormone by growth of one or more follicles. As it grows in size the follicle secretes increasing amounts of a hormone called **oestrogen**. The main effect of oestrogen is to stimulate growth of a new lining in the uterus. This lining consists of a mass of blood vessels and glandular tissue. Oestrogen also stimulates the breasts (mammary glands) to grow the small tubes (ducts) which could eventually carry milk to the nipples.
3. The rising level of oestrogen in the blood acts as a feed-back mechanism: it causes the pituitary to stop producing follicle-stimulating hormone and begin producing another hormone which brings

about ovulation. This usually occurs sometime between days 13 and 15 of the cycle, but it can happen earlier or later than this. Immediately after ovulation a woman is said to be **fertile**, and can conceive—become pregnant—if she has sexual intercourse. Since it is difficult to know exactly when ovulation takes place, and since sperms can live for up to three days in the uterus after intercourse, most women must assume that they are fertile from at least day 11 to day 17 of their cycle (Fig. 1).

4. The part of a follicle left behind in the ovary after ovulation forms a large yellow object called a **corpus luteum** (Fig. 56.3). This acts as a temporary endocrine organ and produces the hormone **progesterone**. Progesterone stimulates further growth of the uterus wall, and it stimulates the breasts to begin developing the glandular tissues which make milk.

5. Meanwhile the ovum has been sucked into a fallopian tube and is moving towards the uterus. If it is not fertilized within thirty-six hours after ovulation the ovum dies.

6. Death of the ovum is soon followed by a gradual reduction in size of the corpus luteum and a reduction in the amount of progesterone it produces. Lack of progesterone brings about another period about 14 days after ovulation.

Menopause

Men can continue producing sperm throughout life but women lose their ability to have children between forty-two and fifty-five. **Menopause** is the technical name for this loss of fertility, but it is commonly known as 'change of life'. At this time ovulation and menstruation stop and the reproductive organs decrease in size.

Fig. 1 (opposite). The **menstrual cycle** is a sequence of events, repeated monthly, in the female reproductive system. These events are controlled by hormones from an ovary and from the pituitary gland. During one 28 day cycle an ovum is released from an ovary and the womb is prepared to receive this ovum should it be fertilized and develop into a baby.

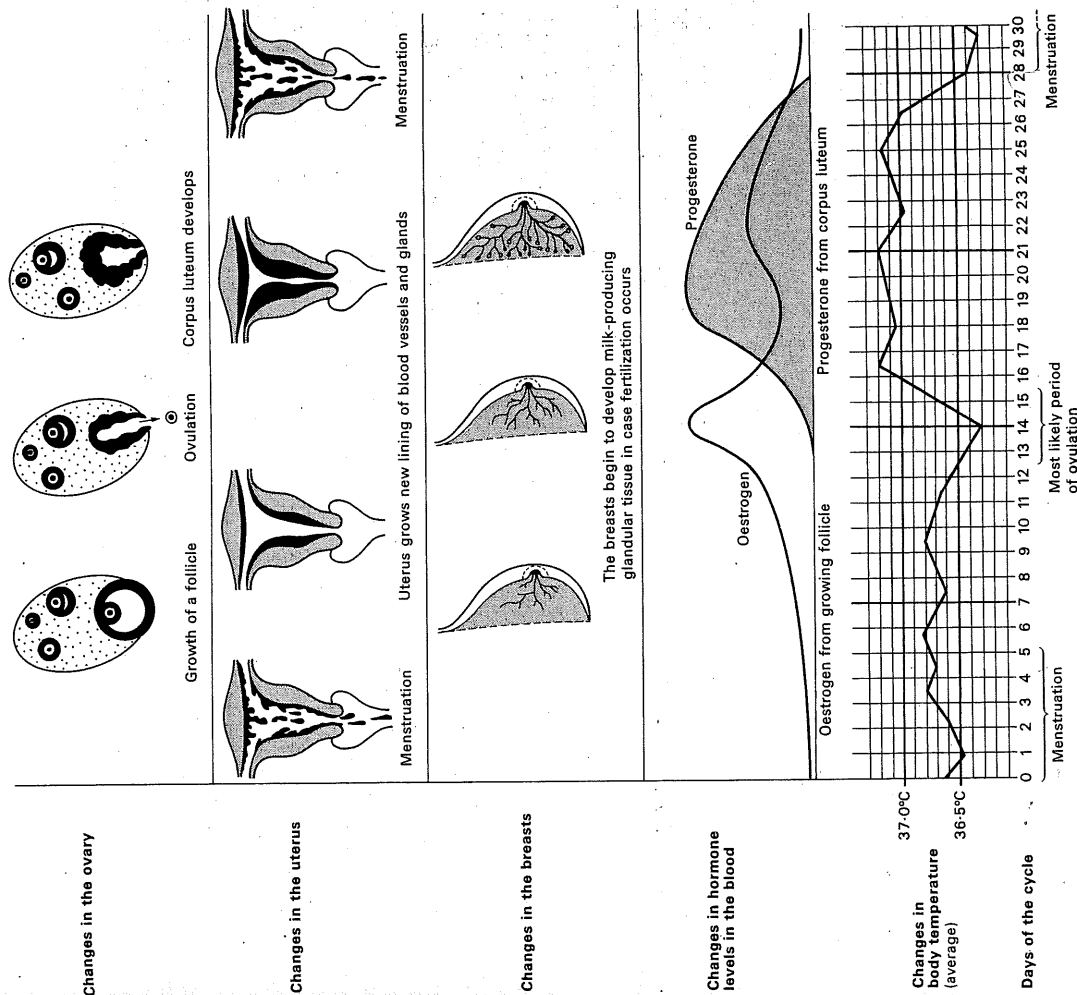


Fig. 1 (opposite). The **menstrual cycle** is a sequence of events, repeated monthly, in the female reproductive system. These events are controlled by hormones from an ovary and from the pituitary gland. During one 28 day cycle an ovum is released from an ovary and the womb is prepared to receive this ovum should it be fertilized and develop into a baby.

Fertilization and implantation

Fertilization

About 5 cm³ of semen, containing up to 300 million sperm, are ejaculated into the female reproductive system during copulation, but only one sperm is needed to fertilize an ovum. A large number of sperm are necessary because many fail to complete the long journey to the area where fertilization occurs.

Sperms swim through the cervix into the uterus and up the fallopian tubes. Usually sperm are sucked into the uterus by muscular contractions of its walls caused by the stimulation of sexual intercourse.

If a ripe ovum is present at the time of copulation fertilization occurs, usually in the upper third of a fallopian tube. The ovum is surrounded by a membrane which a sperm must penetrate. The first sperm to reach the ovum releases a chemical which softens this membrane at the point of contact, allowing the sperm to enter and move into the ovum's cytoplasm. Immediately after penetration by a sperm the membrane undergoes rapid changes which make it thicker and cause it to lift away from the surface of the ovum. It is now called the **fertilization membrane**, and forms a barrier to the entry of other sperm (Fig. 1).

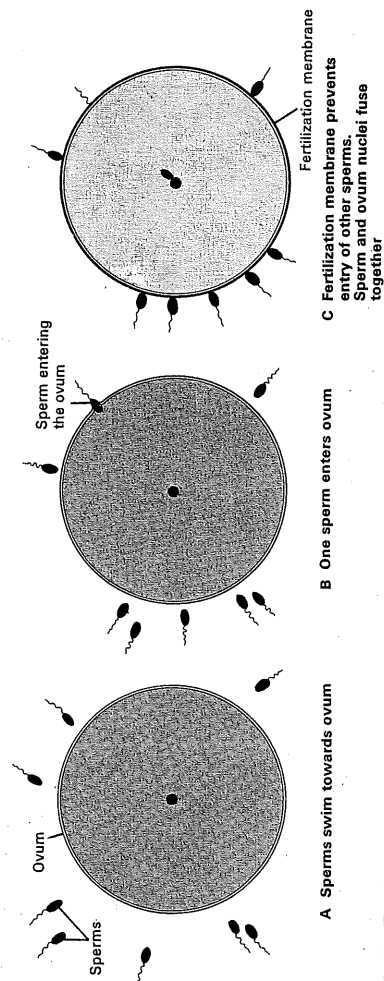


Fig. 1. Fertilization is the fusion of one sperm with an ovum. This event brings together the hereditary characteristics of the mother and father.

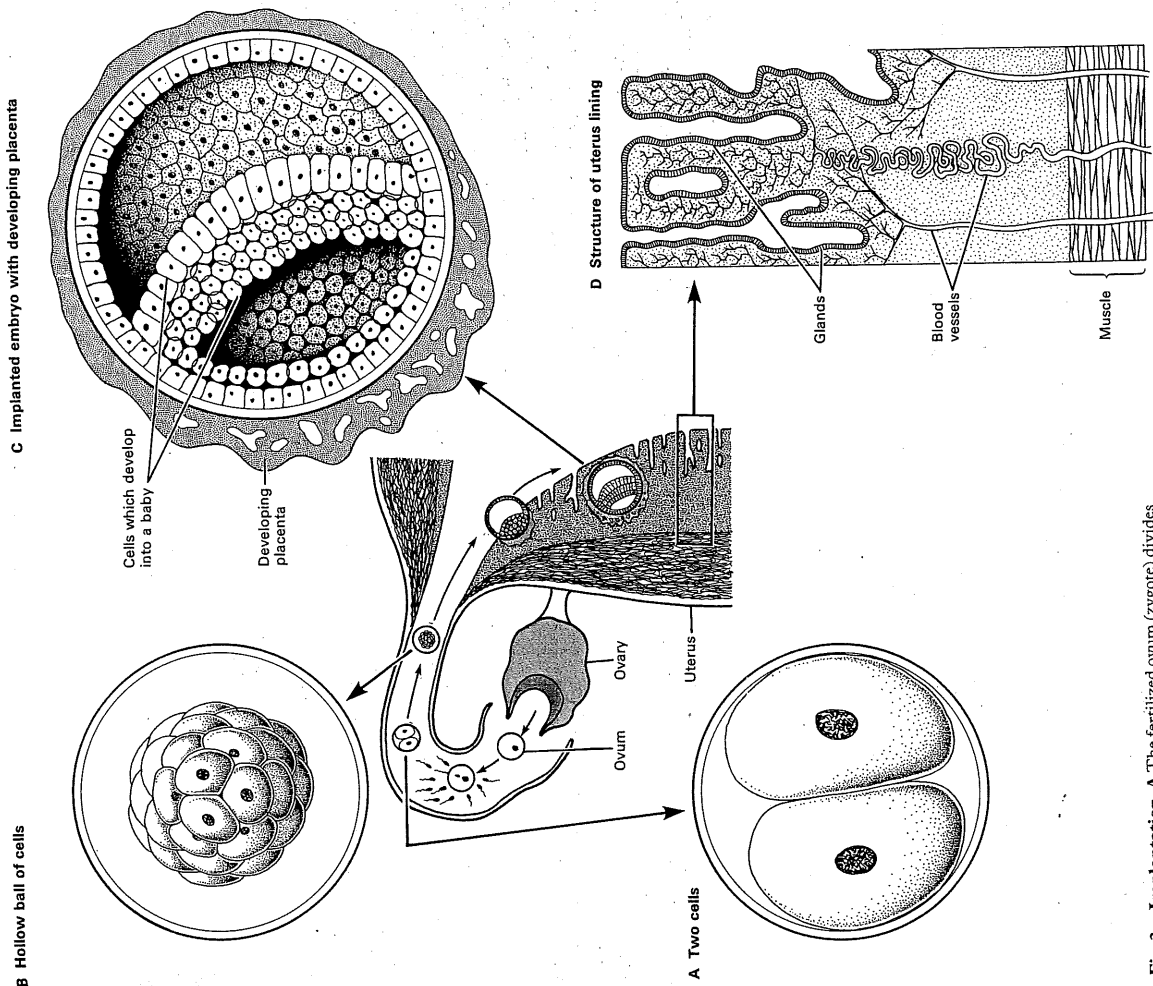
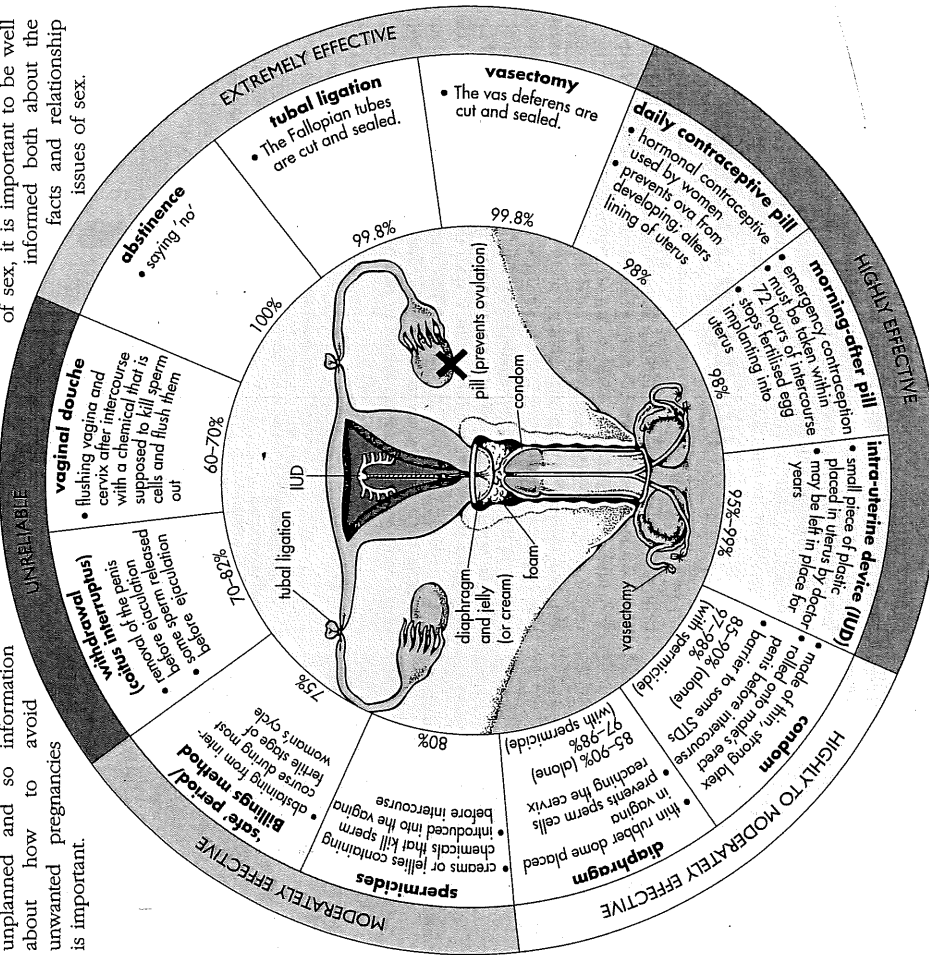


Fig. 2. Implantation. A The fertilized ovum (zygote) divides into two cells. B Cell division continues to make a ball of cells. C The ball becomes hollow and is called an embryo. The embryo sinks into the lining of the uterus. This is called implantation. D Structure of the uterus lining.

To breed or not to breed?

Planning for pregnancy

Conception is another word for the production of a zygote. Techniques that prevent this happening are called **contraception**. **Contraceptives** are devices or substances that prevent fertilisation taking place. Many couples want to choose how many children they have and be able to space their arrivals. It is estimated that, worldwide, half of all pregnancies are unplanned, and so information about how to avoid unwanted pregnancies is important.



Methods of contraception: how do they work and how effective are they?

Classifying contraception

Contraceptives can be classified according to how they function. Contraceptive methods such as the condom and diaphragm work as **barrier** methods because they provide physical barriers to the sperm and egg meeting. A tubal ligation and vasectomy can be thought of as permanent barrier methods. **Chemical** methods include the daily contraceptive pill and spermicides. Lastly, the ‘safe’ period method and abstinence are **natural** methods.

New and improved products

Throughout history, people have tried to find methods that would enable them to have sex but not make babies. In various parts of the world people have tried swallowing tadpoles in spring, using lemons as a ‘diaphragm’ and using pig intestines as condoms. Because there is still no such thing as the perfect contraceptive, many new products are being invented and tested.

For women, some of these include: **transdermal patches** which stick onto the skin and release hormones, **daily vaginal pills** which dissolve into spermicide when inserted into the vagina before intercourse, and the **Filshie clip**, a type of Fallopian tube clamp. In some countries even a female condom is available.

New products for men include: **testosterone injections** or implants to reduce sperm levels, **male anti-fertility vaccines** which regulate sperm and testosterone production, **sperm duct plugs** which involve the injection of liquid plastic into the vas deferens, **chemical sterilisation**, and **gossypol**, a chemical which reduces sperm production.

Activities

Remember

- Which methods of contraception:
 - prevent the sperm from reaching the egg
 - prevent the release of the egg?
- Construct a table which has six columns, with headings for: the type of pregnancy planning, a summary of how it works, who uses it (male or female), suggested advantages, suggested disadvantages, and a prediction of how many pregnancies may occur if 100 couples were to use it.

Think

- After a vasectomy:
 - does a male still produce sperm
 - what does the ejaculate contain?
 - still ovulate
 - menstruate?
- If a contraceptive has a success rate of 95 per cent, that means that the method has a failure rate of 5 per cent. In 100 fertile couples using such a contraceptive, how many pregnancies do you predict might result? Explain.

- Suggest reasons why some contraceptives may be more effective than others.

Research

- Research a method of contraception and prepare a poster on it.
- Find out about four of the following types of contraceptives:
 - combined oral contraceptive pill, the progesterone-only pill (or the mini pill), depo-provera (injectable contraception), hormonal implants, the morning-after pill (MAP), the today sponge, Intra-Levonova IUD, the vaginal ring or female condoms.
- Find out the advantages and disadvantages of each type of contraceptive shown. Present your information in a table.

Create

Design a ‘future’ contraceptive, using your imagination and knowledge of reproductive systems. Decide on how you would scientifically test the safety, effectiveness and popularity of your contraceptive. Produce a brochure that promotes your invention.

Birth control

Birth control allows people to limit the size of their families by either preventing fertilization or implantation, or by limiting sexual intercourse to times when fertilization is unlikely to occur.

Preventing conception

Withdrawal The man withdraws his penis just before ejaculation occurs. The idea is to avoid shedding sperm into the woman but this method is extremely unreliable because many sperm are shed without the man's knowledge some time before ejaculation.

The condom A condom is a sheath of thin strong latex rubber which is unrolled onto the erect penis before intercourse. Sperm are trapped inside the condom and so do not enter the woman. This method is quite reliable, especially if used in conjunction with spermicides.

Spermicides These consist of a jelly, cream, or foam containing chemicals which kill sperm. Spermicides are inserted into the vagina before intercourse but, since they are unreliable on their own, they should be used in conjunction with a condom or diaphragm.

Diaphragm This is a dome-shaped disc of thin rubber which is kept in shape by a spring around the rim. A diaphragm is very reliable provided the woman obtains one which exactly fits her cervix. It must be smeared with a spermicide before fitting and it should be fitted half an hour before intercourse. The diaphragm should not be removed until six to eight hours after intercourse.

Intra-uterine device (IUD) An IUD is a piece of flexible plastic in the form of a loop or coil which is inserted into the womb. It must be fitted by a doctor, and regular medical checks are necessary to ensure that the device remains in place and does not irritate the lining of the womb. This method is very reliable. It appears to work by preventing a fertilized ovum from becoming implanted in the womb.

Sterilization This involves a surgical operation. In men the operation is called **vasectomy**,

and involves cutting, tying, or blocking the tubes which carry sperm from the testes to the penis. In women the operation involves cutting, tying, or blocking the fallopian tubes. The operation is effective immediately in women, but men can still produce live sperm for up to six months afterwards.

Contraceptive pills These contain one or more female hormones. They prevent ovulation, stop the mechanism which transports ova along the fallopian tubes, and cause mucus in the entrance to the womb to become sticky so that sperm are unlikely to swim through. Pills are almost 100 per cent reliable when taken according to instructions. But they must not be taken by women with liver diseases or diabetes, and in rare instances they have been known to cause blood clots. There is evidence that women who smoke and take contraceptive pills are more likely than non-smokers to develop blood clots, migraine headaches, and allergies to certain foods.

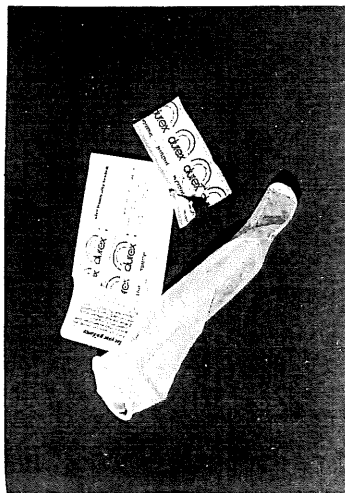
Avoiding conception

The rhythm, or mucothermic, method This method involves avoiding intercourse during the time each month when the woman is fertile. This time can be discovered in two ways. As soon as her menstrual bleeding stops the woman must make daily observations to discover when a thin clear mucus is discharged from the entrance to her vagina. In addition she must record her temperature very carefully every day and note when it goes up by 0.1°C to 0.5°C and stays up for several days. Ovulation occurs soon after the mucus appears and it is followed, within two or three days, by a slight rise in temperature. Consequently intercourse should be avoided after the mucus appears and should not start again until at least three days after the temperature rise occurs.

The temperature rise at ovulation is very small indeed. Consequently this method is only reliable when an accurate clinical thermometer is used correctly, and very careful records are kept.



B Spermicides are chemicals which kill sperm. They are produced in tablet form (pessaries) or as a jelly. Both types are inserted into the vagina before intercourse (a plunger is used to insert the jelly).



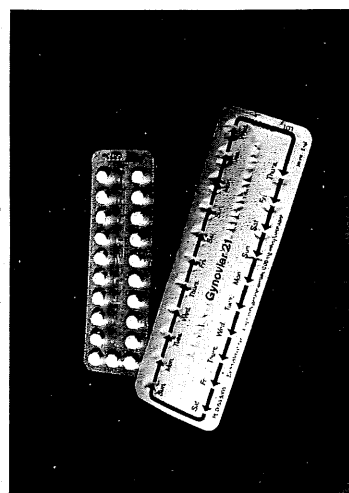
A A condom is a sheath which is unrolled over the penis before intercourse.



C A diaphragm is a dome-shaped disc which is fitted over the cervix before intercourse.



D Intra-uterine devices are inserted into the womb.



E Contraceptive pills contain one or more female hormones which, among other things, prevent ovulation. A pill is taken every day for 21 days, then they are discontinued for the next seven days to allow a 'period' (menstruation) to occur.

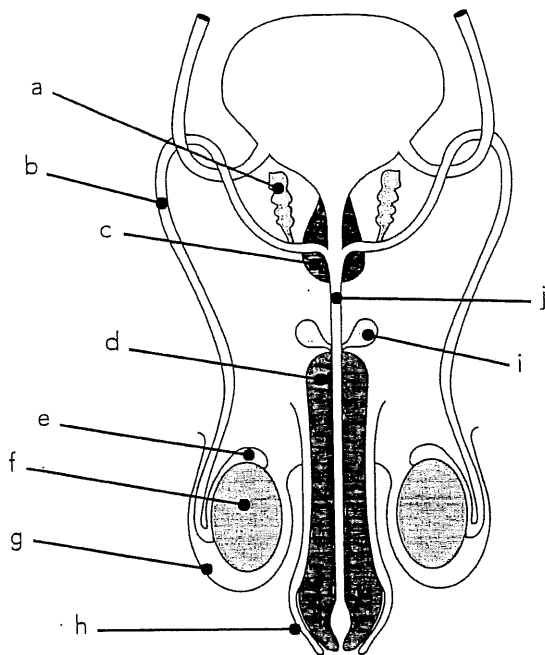


F Pessaries. One of these is inserted into the vagina before intercourse. The pessary dissolves, releasing spermicide chemicals.

SECTION 2 – SHORT ANSWER (50 MARKS)

Answer each question in the space provided.

1. Label the diagram of the male reproductive system below.



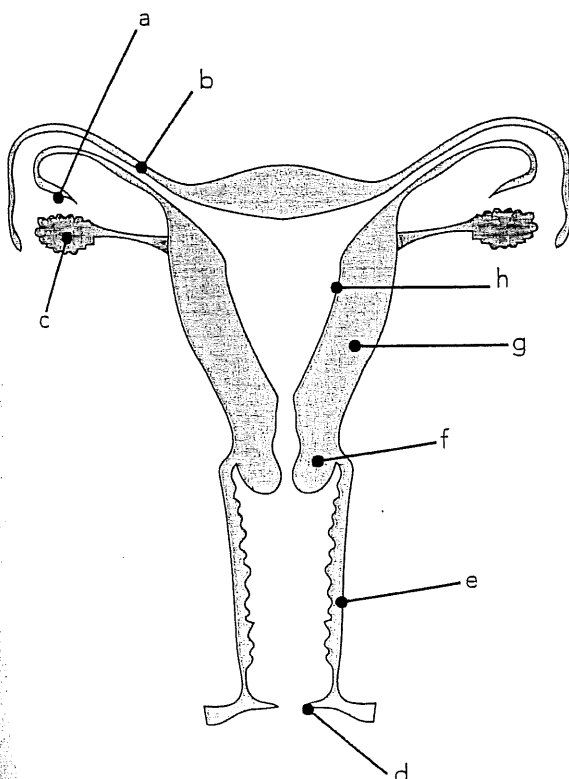
- a) _____
 b) _____
 c) _____
 d) _____
 e) _____
 f) _____
 g) _____
 h) _____
 i) _____
 j) _____

[5 marks]

2. Describe the functions of the male 'accessory glands'.

[1 mark]

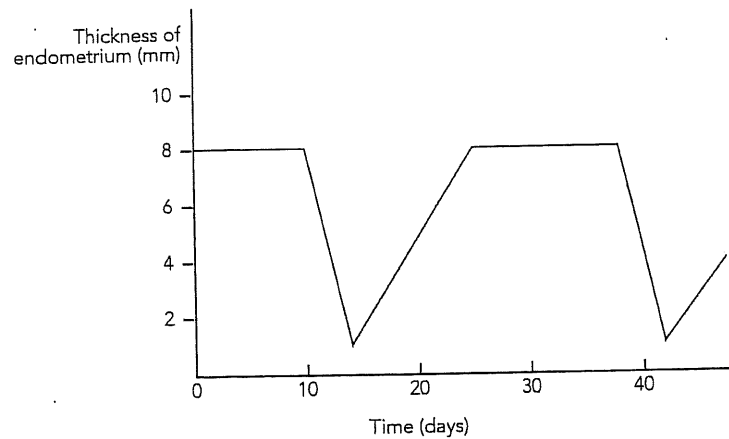
3. Label the diagram of the female reproductive system below:



- a) _____
 b) _____
 c) _____
 d) _____
 e) _____
 f) _____
 g) _____
 h) _____

[4 marks]

The following graph shows the development of a particular adult woman's endometrium over the time shown. Use this graph to answer questions 7 and 8.



6. On which days shown on the graph did menstruation begin:
- day 14 and day 32
 - day 0 and 28
 - day 10 and day 38
 - day 12 and day 40.
7. Ovulation from either one of her ovaries would probably have occurred around:
- day 24
 - day 14
 - day 0
 - day 10 and day 38.

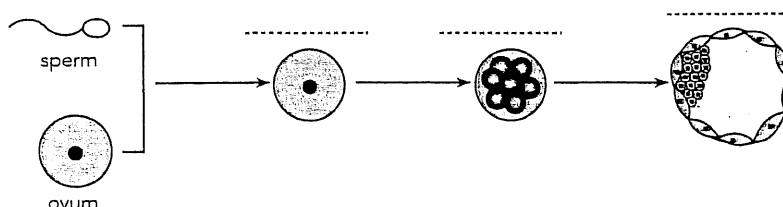
5. (i) Although the blastocyst consists of many more cells than the zygote, it is not very much larger. Explain why this is so.

- (ii) Describe how the cells are produced.

- (iii) Are these cells haploid or diploid? Explain your answer.

6. (i) How many days after fertilisation does implantation normally occur? _____

7. Complete the diagram below:



Complete the following table:

Reproductive system part	Function
cilia	
Cowper's Gland	
Epididymis	
Fallopian Tubes (oviduct)	
foreskin	
Ovary	
Prostate Gland	
Scrotum	
Seminal Vesicle	
Sperm ducts (Vas deferens)	
Testes	
Urethra	
Uterus	
Clitoris	

What is the difference between sexual and asexual reproduction?

What are the benefits of sexual reproduction over asexual?

Complete the table:

	Primary sex characteristics	Secondary sex characteristics
Male		
Female		

Describe what happens between ovulation and implantation. In your answer you should use the words: fallopian tube, cleavage, blastocyst, ovary, ovum, sperm, fertilisation, implantation, zygote, morula, endometrium, nuclei.

Describe how a pregnancy test works.

Create a mindmap or concept map of the menstrual cycle using the following words. Add in the days of the cycle where appropriate.

LH, FSH, Estrogen, Progesterone, Follicular phase, Luteal phase, Ovulation, Endometrium, ovary, uterus, corpus luteum, follicle, ovum, menstruation, proliferative phase, secretory phase, ovarian cycle, uterine cycle.

Menstrual cycle video questions <https://www.youtube.com/watch?v=tOluxtc3Cpw>

1. Where is the endometrium?
2. What are the two parts of the menstrual cycle? _____
3. In a 28-day cycle, roughly when would ovulation occur? _____
4. The follicular phase of the ovarian cycle and the menstrual and proliferative phases of the uterine cycle are _____-ovulatory
5. The luteal phase of the ovarian cycle and the secretory phase of the uterine cycle are _____-ovulatory

Ovarian cycle

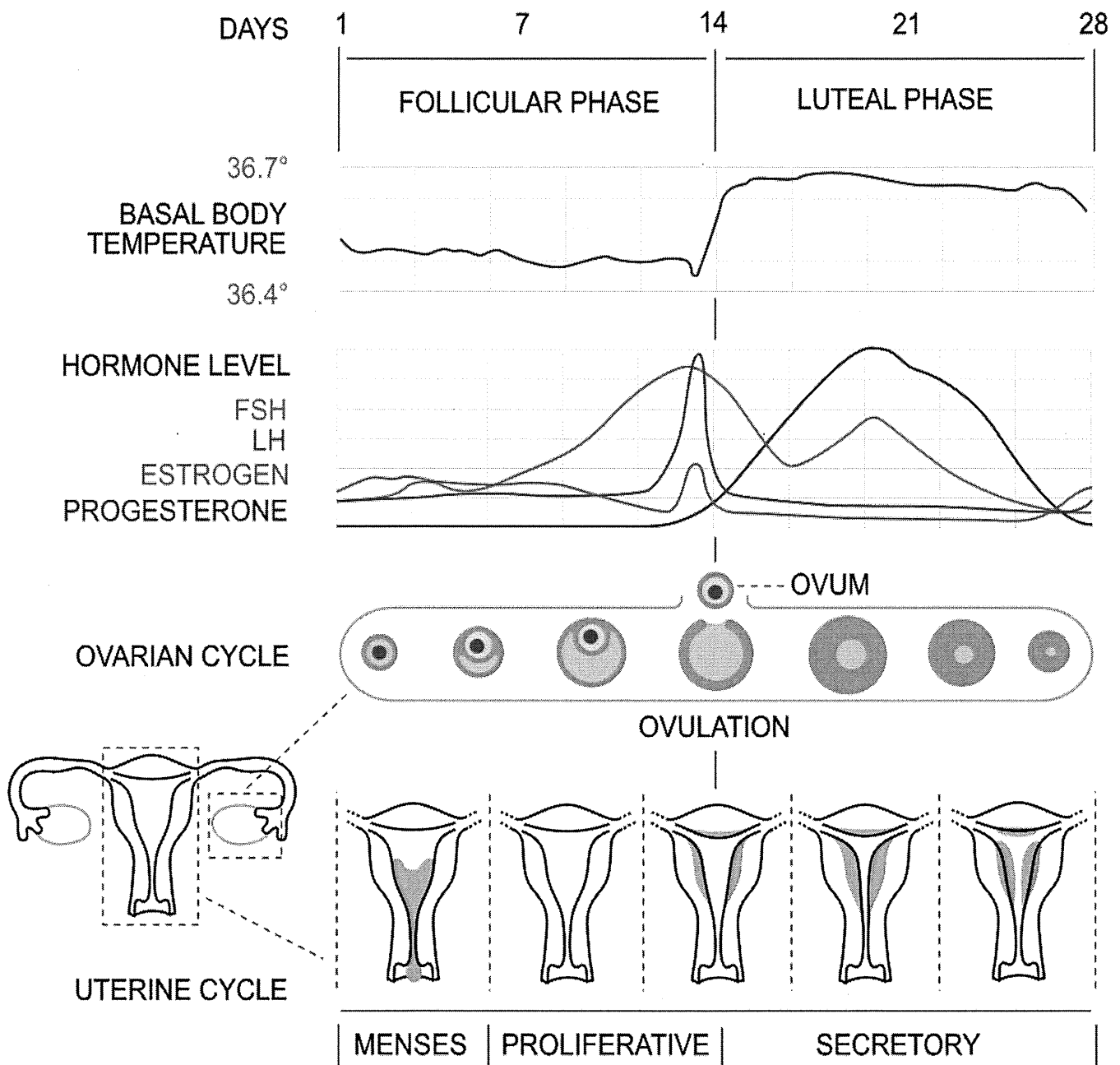
6. Each 'follicle' contains one _____ which is an immature egg
7. One follicle will continue to grow and secrete _____
8. This creates positive feedback to release a surge of _____ and _____ which causes _____
9. Ovulation is when the _____ ruptures and the _____ is released

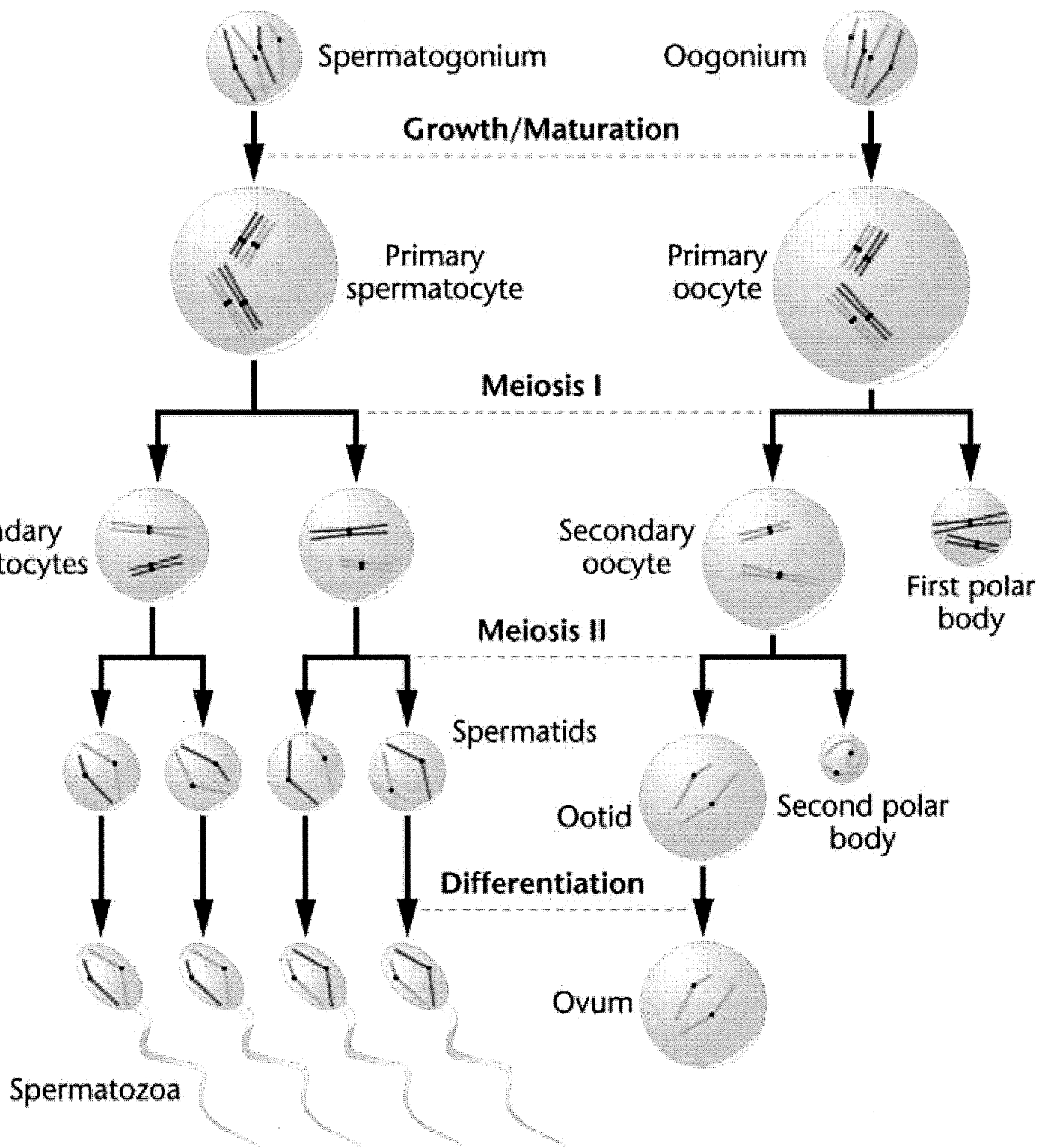
Uterine cycle

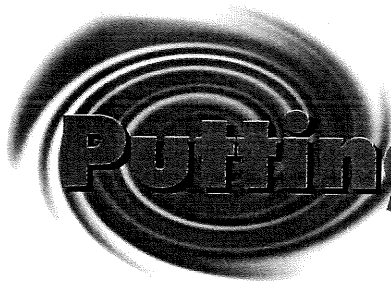
10. This starts of with the _____ phase (bleeding)
11. The average menstrual phase lasts _____ days
12. In the proliferative phase there is high levels of _____
13. This causes the walls of the uterus (_____) to become _____
14. The highest chance of fertilisation is between days _____
15. After ovulation, the follicle becomes a _____
16. In the luteal phase, there is more _____ than _____
17. Progesterone is the main hormone in the _____ (after ovulation). This phase is to make the uterus an ideal environment for implantation.

Recap:

18. The _____ phase may vary in length but the _____ phase is always 14 days







Putting it all together

Summing up

Copy and complete the statements below to compile a summary of this unit. The missing words can be found in the list below.

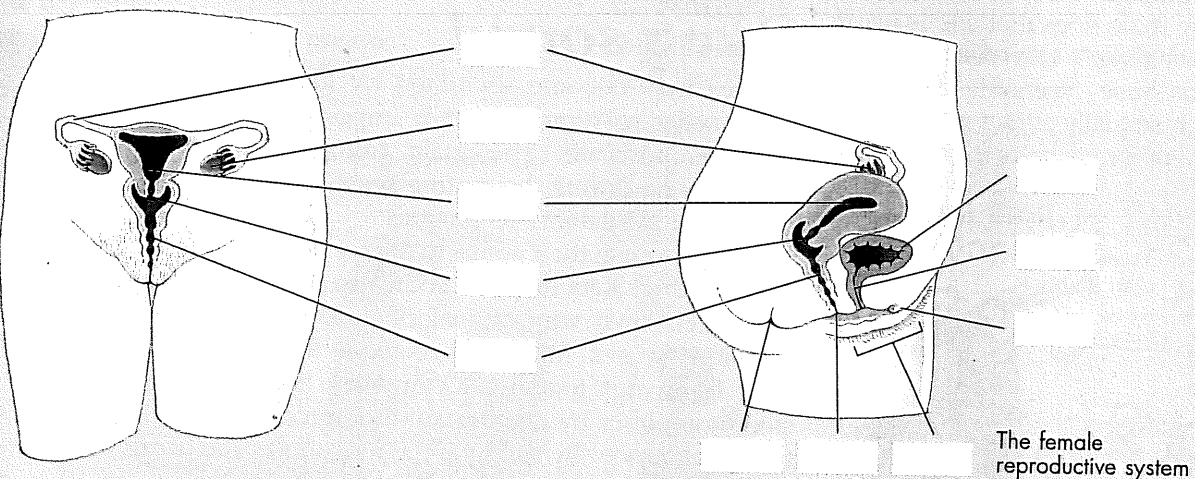
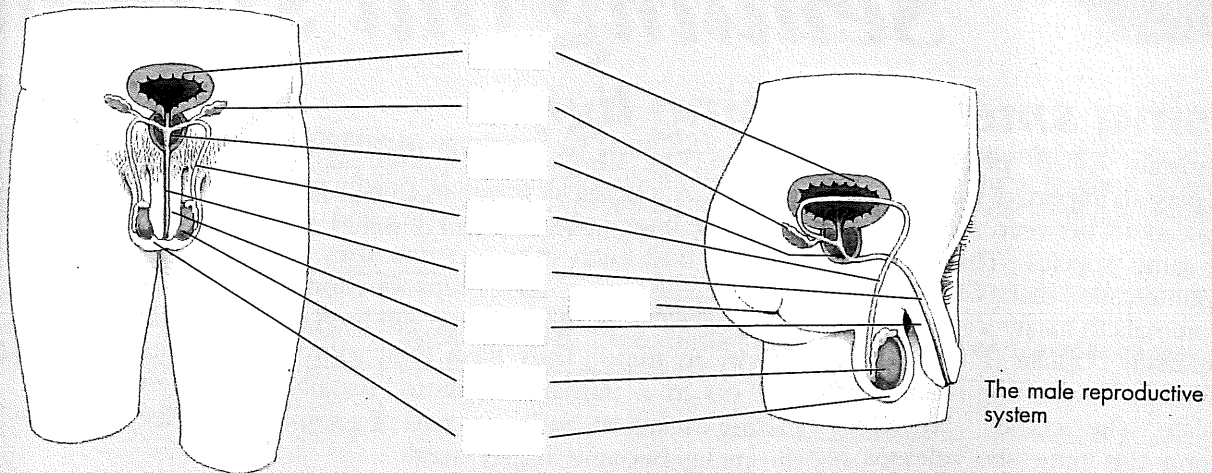
1. _____ reproduction does not require the sex cells of another organism. Examples include vegetative propagation, parthenogenesis, _____, _____ and regeneration. The scientific name for the development of new individuals from an unfertilised egg is _____.
2. Sexual _____ occurs when offspring result from the joining together (_____) of a male reproductive cell and a female reproductive cell. These special reproductive cells are called _____.
3. The joining of a sperm and an egg cell is called _____. The cell resulting from this is called a _____.
4. In some animals, fertilisation occurs outside the female's body. This is called _____. fertilisation. In other animals, for example those that live and breed on land, _____ fertilisation occurs.
5. Human sperm travels through the _____ from the testes to the urethra.
6. Human ova are produced in the _____ whereas sperm are produced in the _____.
7. In humans, the male's _____ is placed in the female's vagina during sexual intercourse. _____ form the tunnel in which the sperm and the egg meet, and hence the site of fertilisation.
8. The _____ is the site of implantation of the embryo and the 'home' of the developing fetus.
9. Changes that you may be currently aware of are indications that you are becoming an adult. These physical changes are called _____.
10. The release of the ovum from the ovary is called _____.
11. The monthly discharge or shedding of the uterus lining is called _____ or a period. A _____ cycle is the time from the first day of one period to the first day of the next.
12. The fetus obtains its nutrients and oxygen through a special organ called the _____. The total time spent in the uterus is often called the _____ period.
13. There are a variety of different strategies used by organisms to increase the chances of fertilisation occurring. Due to _____ factors and predators, not all fertilised eggs survive.

Word list

environmental	binary fission	menstrual	penis	testicles
placenta	asexual	reproduction	menstruation	vas deferens
budding	fusion	fertilisation	ovulation	parthenogenesis
gestation	gametes	zygote	ovaries	Fallopian tubes
puberty	internal	external	uterus	

Looking back

1. Copy the diagrams below and add labels. Write one function of at least two parts in each diagram.



2. Design a calendar of the menstrual cycle and then outline the events that occur at each stage on your calendar.
3. Match the contraceptives with the way they prevent conception and their effectiveness.

Contraceptive	How it prevents conception	Effectiveness
condom with spermicide	prevents ova from developing	extremely effective
diaphragm without spermicide	the Fallopian tubes or vas deferens are cut and sealed	unreliable
daily contraceptive pill	keeps sperm and semen from entering the woman's vagina after ejaculation	highly effective
surgical: vasectomy and tubal ligation	removal of male's penis from the vagina before ejaculation	highly effective
coitus interruptus	prevents sperm cells from reaching the cervix	moderately effective

4. Unscramble the following types of asexual reproduction
taevvegeti gatponproai narybi sfionis gatieneoner sheneipartognes
5. Summarise the disadvantages and advantages of sexual and asexual reproduction.
6. Invent, design and make your own creature. Describe its courting and mating behaviour, and give details about the way it reproduces.